

Lecture 11: The Derivative of $\ln x$

GOAL: Differentiate $\ln x$ and introduce the technique of Logarithmic Differentiation. Finish the proof of the Power Rule.

Summary:

$$[\ln x]' = \frac{1}{x} \quad [\log_b x]' = \frac{1}{x \ln b}$$

$$[\ln f(x)]' = \frac{f'}{f} \quad x^p = e^{p \ln x}$$

Example 1 Find y' if $y = \frac{490t}{(t^2+7)^3}$

$$[490t]' = 490$$

$$[(t^2+7)^3]' = 3(t^2+7)^2 \cdot 2t = 6t(t^2+7)^2$$

$$y' = \frac{490(t^2+7)^3 - 490t(6t)(t^2+7)^2}{(t^2+7)^6}$$

Goal Develop a tool to mitigate this issue

Derivative of $\ln x$

$$[e^{f(x)}]' = e^{f(x)} \cdot f'(x)$$

We can now use this to find $\frac{d}{dx}(\ln x)$

$$f(x) = \ln x$$

$$e^{f(x)} = e^{\ln x}$$

$$\frac{d}{dx}(e^{f(x)}) = \frac{d}{dx}(x)$$

$$e^{f(x)} \cdot f'(x) = 1$$

$$f'(x) = \frac{1}{e^{f(x)}} = \frac{1}{e^{\ln x}} = \frac{1}{x}$$

Theorem (1) $\frac{d}{dx}(\ln x) = \frac{1}{x}$ ✓

(2) If f is a diff. fun, then

$$\frac{d}{dx}(\ln(\boxed{f(x)})) = \frac{1}{f(x)} \cdot f'(x) = \frac{f'(x)}{f(x)}$$

In the chain rule form

$$g(x) = f(x)$$

$$f(-) = \ln(-)$$

Ex 2 $y = \ln(x^2 - 1)$

$$y' = \frac{d}{d(x^2-1)}(\ln(x^2-1)) \cdot \frac{d}{dx}(x^2-1)$$

$$= \frac{1}{x^2-1} \cdot 2x = \frac{2x}{x^2-1}$$

METHOD: $y = \ln(x^2-1) = \ln((x-1)(x+1))$

$$y = \ln(x-1) + \ln(x+1)$$

$$y' = \frac{1}{x-1}(1) + \frac{1}{x+1}(1) = \left[\frac{1}{x-1} + \frac{1}{x+1} \right]$$

$$= \frac{x+1 + (x-1)}{(x-1)(x+1)} = \frac{2x}{x^2-1}$$

Ex 3 $y = \ln\left(\sqrt{\frac{x+1}{x^2+4}}\right) = \ln\left[\left(\frac{x+1}{x^2+4}\right)^{\frac{1}{2}}\right]$

$$= \frac{1}{2} \ln\left[\frac{x+1}{x^2+4}\right] = \frac{1}{2} \left(\ln(x+1) - \ln(x^2+4) \right)$$

$$y = \frac{1}{2} \ln(x+1) - \frac{1}{2} \ln(x^2+4)$$

$$y' = \frac{1}{2} \cdot \frac{1}{x+1} \cdot 1 - \frac{1}{2} \cdot \frac{1}{x^2+4} \cdot 2x$$

$$= \left[\frac{1}{2x+2} - \frac{x}{x^2+4} \right]$$

Ex 4 $y = \log x = \log_{10} x$ Change of Base Formula $\frac{\ln x}{\ln 10}$

$$y' = \frac{1}{\ln 10} [\ln x]' = \frac{1}{\ln 10} \cdot \frac{1}{x} = \frac{1}{x \ln 10}$$

In general, for a base $b > 0$

$$\frac{d}{dx} (\log_b x) = \frac{1}{x \ln b}$$

Logarithmic Differentiation Recall if $y = f(x)$

$$[\ln(y)]' = [\ln(f(x))]' = \frac{1}{f(x)} \cdot f'(x) = \frac{y'}{y}$$

We can use this to our advantage

Example 5 Find y' if $y = \frac{490t}{(t^2+7)^3}$

$$y = \frac{490t}{(t^2+7)^3}$$

$$\ln(y) = \ln\left(\frac{490t}{(t^2+7)^3}\right)$$

$$\ln(y) = \ln(490t) - \sqrt{\ln[(t^2+7)^3]}$$

$$\ln(y) = \ln 490 + \ln t - 3 \ln(t^2+7)$$

$$[\ln(y)]' = [\ln 490 + \ln t - 3 \ln(t^2+7)]'$$

$$\frac{y'}{y} = \frac{1}{t} - 3 \cdot \frac{1}{t^2+7} \cdot 2t$$

$$\frac{y'}{y} = \frac{1}{t} - \frac{6t}{t^2+7}$$

$$y' = y \left[\frac{1}{t} - \frac{6t}{t^2+7} \right]$$

$$y = \frac{490t}{(t^2+7)^3}$$

$$y' = \frac{490t}{(t^2+7)^3} \left[\frac{1}{t} - \frac{6t}{t^2+7} \right]$$

Example 6 Find y' if $y = (2x+1)^5 (x^4-3)^6$

$$\ln y = \ln[(2x+1)^5 (x^4-3)^6]$$

$$= \sqrt{\ln[(2x+1)^5]} + \sqrt{\ln[(x^4-3)^6]}$$

$$\ln y = 5 \ln(2x+1) + 6 \ln(x^4-3)$$

$$\frac{y'}{y} = 5 \cdot \frac{1}{2x+1} \cdot 2 + 6 \cdot \frac{1}{x^4-3} \cdot 4x^3$$

$$\frac{y'}{y} = \frac{10}{2x+1} + \frac{24x^3}{x^4-3}$$

$$y' = y \left[\frac{10}{2x+1} + \frac{24x^3}{x^4-3} \right]$$

$$y' = (2x+1)^5 (x^4-3)^6 \left[\frac{10}{2x+1} + \frac{24x^3}{x^4-3} \right]$$

Example 7 Find y' if $y = x^x$

$$\ln y = \ln(x^x)$$

$$\ln y = x \ln(x)$$

$$\frac{y'}{y} = 1 \cdot \ln(x) + x \cdot \frac{1}{x}$$

$$\frac{y'}{y} = \ln x + 1$$

$$y' = y [\ln x + 1]$$

$$y' = x^x [\ln x + 1]$$

Ex 8 Find y' if $y = (x^2 - 8x + 2)\sqrt{25 - x^2}$

$$\log(a+b) \neq \log a + \log b \quad \ln y = \ln [(x^2 - 8x + 2)\sqrt{25 - x^2}]$$

$$\ln y = \ln(x^2 - 8x + 2) + \ln \sqrt{25 - x^2}$$

$$\ln y = \ln(x^2 - 8x + 2) + \frac{1}{2} \ln(25 - x^2)$$

$$\frac{y'}{y} = \frac{1}{x^2 - 8x + 2} (2x - 8) + \frac{1}{2} \cdot \frac{1}{25 - x^2} (-2x)$$

$$\frac{y'}{y} = \frac{2x - 8}{x^2 - 8x + 2} + \frac{-x}{25 - x^2}$$

$$y' = y \left[\frac{2x-8}{x^2-8x+2} - \frac{x}{25-x^2} \right]$$

$$y' = (x^2-8x+2)\sqrt{25-x^2} \left[\frac{2x-8}{x^2-8x+2} - \frac{x}{25-x^2} \right]$$

Derivative of x^p

If p is rational ✓

If p is irrational, x^p is only defined when $x > 0$

$$[x^p]' = [e^{\ln(x^p)}]' = [e^{p \ln(x)}]'$$

$$= \underbrace{e^{p \ln(x)}}_{x^p} \cdot p \cdot \frac{1}{x}$$

$$= x^p \cdot p \cdot \frac{1}{x} = p x^{p-1}$$

$$y = \frac{(x-2)^5 \sqrt{x^2+1}}{(x+3)^7}$$

$$\ln y = 5 \ln(x-2) + \frac{1}{2} \ln(x^2+1) - 7 \ln(x+3)$$

$$\frac{y'}{y} = 5 \cdot \frac{1}{x-2} + \frac{1}{2} \cdot \frac{1}{x^2+1} \cdot 2x - 7 \cdot \frac{1}{x+3}$$